Assignment **(05/03/2026)** – Rajath S

Assignment Name : Component Hunt

Description : Break a popular website into components and sketch the structure.

I have Chosen the website X – twitter - https://x.com

1. The main wireframe mockup of the twitter website

X (Twitter) — UI mockup

2. The components of that website – the component tree

Component tree

- AppShell
  Root layout, routing
  - LeftNav
    Primary navigation rail
    - Logo
      Brand mark + home link
    - NavItem[ ]
      Home · Explore · Notifs…
    - TweetButton
      Opens compose modal
    - UserCard
      Avatar + handle + menu

- MainFeed
  Center scroll column
  - FeedHeader
    For You / Following tabs
  - ComposeTweet
    Input + media toolbar
  - TweetList
    Virtualized infinite list
    - TweetCard
      Individual post unit
      - Avatar
        Profile photo
      - TweetHeader
        Name · handle · time
      - TweetBody
        Text + optional media
      - ActionBar
        Reply · RT · Like · Share
- RightSidebar
  Discovery panel
  - SearchBar
    Full-text search input
  - TrendingWidget
    What's happening topics
  - SuggestWidget
    Who to follow

Shell — AppShell is the single root that owns the three-column layout and handles routing.

Navigation — LeftNav holds four children: a Logo, a repeated NavItem[] list, the TweetButton that opens the compose modal, and UserCard pinned to the bottom.

Feed — MainFeed stacks three things vertically: a sticky FeedHeader with the For You / Following tabs, a ComposeTweet bar, and a TweetList that renders an infinite virtualized scroll.

Tweet — TweetCard is the most compositional unit — it's just a shell for Avatar, TweetHeader, TweetBody, and ActionBar. Each of those is independently reusable (the Avatar appears in composing, replies, DMs, etc.).

Sidebar — RightSidebar stitches three discovery widgets: SearchBar, TrendingWidget, and SuggestWidget. These are slot-filled — the sidebar doesn't know what it renders, just where.

The key insight: **TweetCard is the real unit of reuse.** Everything else is structural scaffolding around it